

Lambda calculus

Exercise sheet 5 (solutions)

Exercise 5.1. Let $ID = \lambda x.x$ and $\Omega = (\lambda x.xx)(\lambda x.xx)$. Reduce the following terms to normal form:
 (a) $(\lambda x.\lambda y.yx) z ID$; (b) $(\lambda x.\lambda y.xy) y$; (c) $TRUE ID \Omega$; (d) $\bar{1} \bar{2} \bar{3}$; (e) $SND(FST(PAIR(PAIR x y) z))$.

Solution.

$$(a) (\lambda x.\lambda y.yx) z ID \longrightarrow_{\beta} (\lambda y.yz) ID \longrightarrow_{\beta} ID z \longrightarrow_{\beta} z$$

$$(b) \text{Rename the inner parameter to } z \text{ so that the argument } y \text{ remains free:}$$

$$(\lambda x.\lambda y.xy) y \equiv_{\alpha} (\lambda x.\lambda z.xz) y \longrightarrow_{\beta} \lambda z.yz$$

$$(c) TRUE ID \Omega \longrightarrow_{\beta}^* ID$$

□

Exercise 5.2. Define as lambda terms: (a) disjunction; (b) implication; (c) exclusive disjunction.

Solution. Disjunction returns true if p is true, and otherwise returns q . Implication returns q if p is true, and otherwise returns true. Exclusive disjunction negates q if p is true, and otherwise returns q .

$$\begin{aligned} OR &= \lambda p.\lambda q.p p q, \\ IMPLIES &= \lambda p.\lambda q.p q TRUE, \\ XOR &= \lambda p.\lambda q.p (NOT q) q. \end{aligned}$$

We can also express Boolean functions using NOT and AND, e.g.

$$OR AB \quad \text{is equivalent to} \quad NOT(AND(NOTA)(NOTB))$$

□

Exercise 5.3. Define the following functions for Church numerals: (a) zero test: $\bar{0} \mapsto \text{TRUE}$, $\bar{0} \neq \bar{n} \mapsto \text{FALSE}$; (b) parity test: $\overline{2k} \mapsto \text{TRUE}$, $\overline{2k+1} \mapsto \text{FALSE}$; (c) exponentiation: $\bar{n}, \bar{m} \mapsto \overline{n^m}$.

Solution. Recall that a numeral $\bar{n}$ takes a function f and an argument x and iterates f on x exactly n times.

- (a) Pass the function $\lambda x.\text{FALSE}$ (returns FALSE on any input) and the argument TRUE to $\bar{n}$. Zero iterations leave TRUE unchanged; any positive number of iterations returns FALSE:

$$\text{ISZERO} = \lambda n.n (\lambda x.\text{FALSE}) \text{TRUE}.$$

□